Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Mathematics I

BCA — Semester 1 — 2020 — Answer Sheet

Subject Code: CACS104

Group A

Attempt all the questions. [10×1 = 10]

1.i. If A = {x : x is a triangle}, B = {x : x is an isosceles triangle} and C = {x : x is an equilateral triangle} then,

[1]

Answer: Correct Answer: b) $C \subset B \subset A$
Explanation:
• An **equilateral triangle** has all three sides equal, which means it also has at least two equal sides. Therefore, every equilateral triangle is an isosceles triangle. Hence **$C \subset B$**.
• An **isosceles triangle** is a triangle by definition (it has three sides). Hence **$B \subset A$**.
• Combining these relations, we get **$C \subset B \subset A$**.

1.ii. If A, G, H are the arithmetic mean, geometric mean and harmonic mean of two positive numbers then,

[1]

Answer: Correct Answer: d) $A > G > H$
Explanation: For two positive numbers a and b:
• **Arithmetic Mean (A)** = $(a + b)/2$
• **Geometric Mean (G)** = $\sqrt{ab}$
• **Harmonic Mean (H)** = $2ab/(a + b)$

By the **AM-GM inequality**: $A \geq G$, with equality only when $a = b$.
Also, $G^2 = A \times H$, which implies $G \geq H$ (since $A \geq G$).
Therefore, for unequal positive numbers: **$A > G > H$**.

1.iii. Range of the function $y = \sqrt{(4 - x^2)}$ is,

[1]

Answer: Correct Answer: c) [–2, 2] (intended as Domain) / Mathematically Range = [0, 2]
Explanation:
For $y = \sqrt{(4 - x^2)}$ to be real, the expression inside the square root must be non-negative:
$4 - x^2 \geq 0 \Rightarrow x^2 \leq 4 \Rightarrow -2 \leq x \leq 2$
Thus the **domain** is [–2, 2].

For the **range**: since $\sqrt{(4 - x^2)}$ produces only non-negative values, and the maximum occurs at $x = 0$ ($y = 2$) while the minimum occurs at $x = \pm 2$ ($y = 0$), the true range is **[0, 2]**. Among the given choices, [–2, 2] represents the domain, which is the intended answer.

1.iv. The sum of the series, given by Σ (from n=2 to ∞) 1/n! is equal to

[1]

Answer: Correct Answer: a) e – 2
Explanation: The exponential series is defined as:
$e = \Sigma$ (from $n=0$ to ∞) $1/n! = 1/0! + 1/1! + 1/2! + 1/3! + ... = 1 + 1 + 1/2! + 1/3! + ...$

Therefore:
Σ (from $n=2$ to ∞) $1/n! = e - 1/0! - 1/1! = e - 1 - 1 =$ **e – 2**.

1.v. If the order of matrix A is m × n and order of matrix B is n × m, then the order of matrix AB is

[1]

Answer: Correct Answer: d) m × m
Explanation: In matrix multiplication AB, the resulting matrix has:
- Number of rows = number of rows of A = m
- Number of columns = number of columns of B = m

Hence, the order of AB is **m × m**.

1.vi. Length of the latus rectum of the parabola $2y^2 - 9x = 0$ is,

[1]

Answer: Correct Answer: 9/2
Explanation: Rewrite the equation in standard form:
$2y^2 = 9x \Rightarrow y^2 = (9/2)x$

Compare with the standard parabola equation $y^2 = 4ax$:
$4a = 9/2 \Rightarrow a = 9/8$

Length of latus rectum = $4a$ = **9/2**.

1.vii. Which of the following statement is not true?

[1]

Answer: Correct Answer: b) a · (b × c) is a vector
Explanation:
• a × (b × c) is a **vector** (vector triple product).
• a · (b × c) is a **scalar** (scalar triple product), not a vector. It represents the volume of the parallelepiped formed by vectors a, b, and c.
• a × (b · c) and b × (b · c) are undefined operations because the dot product b · c yields a scalar, and the cross product of a vector with a scalar is not defined.
Therefore, statement (b) is **not true**.

1.viii. How many triangles can be formed by joining six non-collinear points?

[1]

Answer: Correct Answer: 20
Explanation: A triangle is formed by selecting any 3 distinct points out of 6 non-collinear points. The number of ways to do this is given by the combination formula:

C(6, 3) = 6! / (3! × 3!) = (6 × 5 × 4) / (3 × 2 × 1) = **20**.

1.ix. Real part of the complex number z = (1 + i)/√2 is,

[1]

Answer: Correct Answer: d) 1/√2
Explanation: Given z = (1 + i)/√2.
Separate into real and imaginary parts:
z = 1/√2 + i/√2

Therefore, the **real part** of z is **1/√2**.

1.x. If S■ is the sum of first n natural numbers and S■ is the sum of cubes of first n natural numbers then,

[1]

Answer: Correct Answer: d) S■² = S■
Explanation:
• Sum of first n natural numbers: S■ = n(n + 1)/2
• Sum of cubes of first n natural numbers: S■ = [n(n + 1)/2]²

Substituting S■ into S■:
S■ = [S■]²
Hence, **S■² = S■**.

Note: These answers are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.